

06 – Canvas App (for end users)

Architecture slide: Canvas App covers **Request Intake / APPG Review / Feasibility Assessment / Funding Approval / Execution Tracking / Closure**, plus **Validation & Error Handling** and **Export to PDF**.

1. Create the app

1. make.powerapps.com → LAM-TVRMS-DEV → **Solutions** → **TVRMS** → **New** → **App** → **Canvas app**.
2. Name: **TV Request Management**, Format: **Tablet** (works well in Edge on laptop).
3. **Data** pane → Add data → Dataverse tables: TV Requests, Feasibility Options, Funding Approvals, Milestones, Risk Issues, Cost Entries, Request Comments, Request Documents, Status Histories, Departments, Cost Centers, Locations, Research Institutions, App Settings, Users.
4. Settings → turn ON **"Formula-level error management"** and **"Named formulas"**.

Naming convention for controls: `scr` screen, `gal` gallery, `frm` form, `btn` button, `txt` text input, `dd` dropdown, `cmb` combo box, `dp` date picker, `lbl` label, `att` attachment. Example: `btnSubmit`, `galMyRequests`.

2. Global choices used below

Create these as **global choices** (Solution → New → Choice) so formulas are readable: `TVR Status`, `TVR Phase`, `TVR Request Type`, `TVR Priority`. In Power Fx you write them like: `'TVR Status'.Submitted`.

3. App start-up

```
// App.Formulas
gblAppColorPrimary = ColorValue("#3B2BB3");
CurrentUserRecord = LookUp(Users, 'Primary Email' = User().Email);
IsAPPG = !IsEmpty(Filter(LookUp(Teams, 'Team Name' = "TVRMS APPG").'Users (teammembership_association)',
    'Primary Email' = User().Email));
IsAdmin = !IsEmpty(Filter(LookUp(Teams, 'Team Name' = "TVRMS Admins").'Users (teammembership_association)',
    'Primary Email' = User().Email));

// App.OnStart (deep link support: open a request directly from an email link)
If(
    !IsBlank(Param("TVID")),
    Set(varSelectedRequest, LookUp('TV Requests', TVID = Param("TVID")));
    Navigate(scrRequestDetail)
)
```

Every email link looks like: `{AppUrl}?TVID=TV-2026-00015` → opens the request directly.

4. Screens (list)

#	Screen	Who uses it	Purpose
1	<code>scrHome</code>	Everyone	My requests, counters, "New Request" button
2	<code>scrNewRequest</code>	Requester	3-step wizard to create & submit
3	<code>scrRequestDetail</code>	Everyone (content depends on role)	All information about one TVID in tabs
4	<code>scrAPPGQueue</code>	APPG	Work queue of requests to review
5	<code>scrFeasibility</code>	APPG	Compare research institutions / options
6	<code>scrFunding</code>	APPG / CCO	Send for funding, see approval history

#	Screen	Who uses it	Purpose
7	scrExecution	APPG	Milestones, risks/issues, costs, progress
8	scrClosure	APPG	Final outcome, lessons learned, close request

5. Screen 1 – Home (scrHome)

Layout: header bar (logo, title, user name), 4 counter tiles, a filter row, a gallery.

Counter tiles (labels):

```
// lblMyOpen.Text
CountRows(Filter('TV Requests', Requester.'Primary Email' = User().Email,
    !('Request Status' in ['TVR Status'.Closed, 'TVR Status'.Cancelled, 'TVR Status'.Rejected, 'TVR
        Status'.Funding Rejected'])))

// lblNeedsMyAction.Text (requests waiting for my answer)
CountRows(Filter('TV Requests', Requester.'Primary Email' = User().Email,
    'Request Status' = 'TVR Status'.More Info Required'))
```

Gallery galMyRequests.Items:

```
SortByColumns(
    Filter(
        'TV Requests',
        (IsAPPG || Requester.'Primary Email' = User().Email),
        IsBlank(txtSearch.Text) || StartsWith(TVID, txtSearch.Text) || StartsWith('Request Title',
            txtSearch.Text),
        IsBlank(ddStatusFilter.Selected.Value) || 'Request Status' = ddStatusFilter.Selected.Value
    ),
    "createdon", SortOrder.Descending
)
```

(Dataverse security already limits rows; the filter just makes the list cleaner.)

Inside the gallery show: TVID, Title, Status (coloured pill), Required By, Phase. Status colour:

```
// lblStatusPill.Fill
Switch(ThisItem.'Request Status',
    'TVR Status'.More Info Required, ColorValue("#F4B400"),
    'TVR Status'.Rejected, ColorValue("#D93025"),
    'TVR Status'.Funding Rejected, ColorValue("#D93025"),
    'TVR Status'.Closed, ColorValue("#188038"),
    ColorValue("#1A73E8"))
```

galMyRequests.OnSelect: Set(varSelectedRequest, ThisItem); Navigate(scrRequestDetail)

btnNewRequest.OnSelect: NewForm(frmRequest); Set(varStep, 1); Navigate(scrNewRequest)

6. Screen 2 – New Request wizard (`scrNewRequest`)

Use **one Edit Form** `frmRequest` (DataSource = 'TV Requests') and show fields step by step using `varStep`.

Step	Fields	Sample input
1 – Basic	Title*, Request Type*, Previous Request (if Repeat), Frequency (if Recurring), Department*, Location*, Cost Center*, Priority*, Required By*, Estimated Budget*	"Plasma etch uniformity test", New, –, –, Process Engineering, Fremont CA, CC-4100, High, 15-Dec-2026, 42000
2 – Details	Business Justification*, Technical Description*, Expected Outcome*	text
3 – Documents & Submit	Attachment control (technical docs*), other docs, Review summary, Save as Draft / Submit	Etch_Test_Spec_v1.pdf

Show/hide data cards:

```
// DataCard "Previous Request".Visible
ddRequestType.Selected.Value = 'TVR Request Type'.Repeat
// DataCard "Recurrence Frequency".Visible
ddRequestType.Selected.Value = 'TVR Request Type'.Recurring
```

Repeat request – copy data from old TVID (Previous Request combo `cmbPrevious.OnChange`):

```
If(!IsBlank(cmbPrevious.Selected),
    Set(varCopyFrom, cmbPrevious.Selected);
    Notify("Details copied from " & varCopyFrom.TVID & ". Please review.", NotificationType.Information)
)
// then each text input Default = If(!IsBlank(varCopyFrom), varCopyFrom.'Business Justification',
Parent.Default)
```

Attachment control: Dataverse forms don't give a free attachment control, so add a SharePoint/Dataverse form once, copy its Attachments card's control, paste it into `scrNewRequest`, rename it `attTechDocs`. Set `MaxAttachments = 10`, `MaxAttachmentSize = 25` (MB).

Validation + Submit (`btnSubmit.OnSelect`)

```
// 1. Validation (mandatory fields + technical document)
If(
    !frmRequest.Valid,
        Notify("Please fill all fields marked with *", NotificationType.Error),
    CountRows(attTechDocs.Attachments) = 0,
        Notify("Please attach at least one technical specification document.", NotificationType.Error),
    dpRequiredBy.SelectedDate <= Today(),
        Notify("'Required By' date must be a future date.", NotificationType.Error),
    ddRequestType.Selected.Value = 'TVR Request Type'.Repeat && IsBlank(cmbPrevious.Selected),
        Notify("Please select the previous request for a Repeat request.", NotificationType.Error),

    // 2. All good -> save the request as Draft first (gets a TVID)
    Set(varNew,
        Patch('TV Requests',
            If(IsBlank(varDraft), Defaults('TV Requests'), varDraft),
            frmRequest.Updates,
            { Requester: CurrentUserRecord, 'Request Status': 'TVR Status'.Draft, 'Current Phase': 'TVR Phase'.Initiation }
        )
    );
    IfError(varNew, Notify("Could not save: " & FirstError.Message, NotificationType.Error); Exit());

    // 3. Upload each file to SharePoint through flow F09
    ForAll(attTechDocs.Attachments As f,
        'F09-UploadDocument'.Run(varNew.TVID, "Technical Specification", "02-Technical Documents",
            { file: { name: f.Name, contentBytes: f.Value } })
    );
```

```
// 4. Change status to Submitted (this triggers flow F01)
Patch('TV Requests', varNew, { 'Request Status': 'TVR Status'.Submitted, 'Current Phase': 'TVR
Phase'.APPG Review', 'Submitted On': Now(), 'Status Changed On': Now() });
Patch('Status Histories', Defaults('Status Histories'),
  { Name: varNew.TVID & ": Draft → Submitted", 'TV Request': varNew,
    'From Status': 'TVR Status'.Draft, 'To Status': 'TVR Status'.Submitted,
    'Changed On': Now(), Remarks: "Submitted by requester" });

Notify("Request " & varNew.TVID & " submitted successfully.", NotificationType.Success);
Reset(attTechDocs); Set(varDraft, Blank());
Navigate(scrHome)
)
```

`btnSaveDraft.OnSelect` = same Patch with `'Request Status': 'TVR Status'.Draft` and **no** validation except Title. Store result in `varDraft` so the next save updates the same record.

7. Reusable "change status" pattern

Many screens change status. Use one hidden button `btnDoStatusChange` on the detail screen:

```
// btnDoStatusChange.OnSelect (inputs: varNewStatus, varRemarks)
With({ oldStatus: varSelectedRequest.'Request Status' },
  Set(varSelectedRequest,
    Patch('TV Requests', varSelectedRequest,
      { 'Request Status': varNewStatus, 'Status Changed On': Now(),
        'Current Phase': Switch(varNewStatus,
          'TVR Status'.Internal Alignment 1, 'TVR Phase'.Internal Alignment',
          'TVR Status'.Internal Alignment 2, 'TVR Phase'.Internal Alignment',
          'TVR Status'.Pending Funding Approval, 'TVR Phase'.Funding Approval',
          'TVR Status'.Funding Approved, 'TVR Phase'.Funding Approval',
          'TVR Status'.In Execution, 'TVR Phase'.Execution,
          'TVR Status'.On Hold, 'TVR Phase'.Execution,
          'TVR Status'.Execution Completed, 'TVR Phase'.Closure,
          'TVR Status'.Closed, 'TVR Phase'.Closure,
          varSelectedRequest.'Current Phase') }) // other statuses keep the same phase
    );
  Patch('Status Histories', Defaults('Status Histories'),
    { Name: varSelectedRequest.TVID & ": " & Text(oldStatus) & " → " & Text(varNewStatus),
      'TV Request': varSelectedRequest,
      'From Status': oldStatus, 'To Status': varNewStatus,
      'Changed On': Now(), Remarks: varRemarks });
  If(!IsBlank(varRemarks),
    Patch('Request Comments', Defaults('Request Comments'),
      { Comment: varRemarks, 'TV Request': varSelectedRequest, Stage: varSelectedRequest.'Current
        Phase' }));
)
```

Any button then does:

```
Set(varNewStatus, 'TVR Status'.More Info Required);
Set(varRemarks, txtReviewComment.Text);
Select(btnDoStatusChange)
```

Flow **F02** sees the status change and sends the right email.

8. Screen 3 – Request Detail (`scrRequestDetail`)

Top: TVID, Title, Status pill, Phase **progress bar** (6 chevrons; current phase highlighted). Tabs (use a horizontal gallery of tab names + `varTab`):

Tab	Content	Edit allowed for
Summary	Display form of the request	Requester (only in Draft / More Info Required)
Comments	Gallery of Request Comments + add box	Everyone with access
Documents	Gallery of Request Documents (click opens SharePoint link) + upload	Requester, APPG
Feasibility	Options gallery (read-only for requester)	APPG
Funding	Funding Approvals history	APPG (send), CCO (read)
Execution	Milestones, Risks/Issues, Costs	APPG
Closure	Final outcome, lessons learned, reuse	APPG
History	Status Histories, newest first	read only

Action buttons (visible by role and status):

Button	Visible when	Sets status to
Start Review	<code>IsAPPG && 'Request Status' = Submitted</code>	Under APPG Review (and APPG Owner = me)
Request More Info	<code>IsAPPG && 'Request Status' = Under APPG Review</code>	More Info Required (comment mandatory)
Reject	<code>IsAPPG && 'Request Status' in [Under APPG Review, Internal Alignment 1]</code>	Rejected (reason mandatory)
Resubmit	<code>Requester = me && 'Request Status' = More Info Required</code>	Under APPG Review
Move to Internal Alignment	<code>IsAPPG && 'Request Status' = Under APPG Review</code>	Internal Alignment 1
Confirm Option (Alignment 2)	<code>IsAPPG && 'Request Status' = Internal Alignment 1 && an option is selected</code>	Internal Alignment 2
Send for Funding Approval	<code>IsAPPG && 'Request Status' = Internal Alignment 2</code>	Pending Funding Approval (flow F04 starts)
Start Execution	<code>IsAPPG && 'Request Status' = Funding Approved</code>	In Execution
Put On Hold / Resume	<code>IsAPPG</code>	On Hold / In Execution
Complete Execution	<code>IsAPPG && 'Request Status' = In Execution</code>	Execution Completed
Close Request	<code>IsAPPG && 'Request Status' = Execution Completed && Final Report uploaded</code>	Closed (flow F07)
Cancel	<code>(Requester = me && 'Request Status' in [Draft, Submitted, More Info Required]) IsAPPG</code>	Cancelled
Export to PDF	everyone	calls flow F06

Example "Reject" with mandatory reason:

```
// btnReject.OnSelect
If(IsBlank(txtReviewComment.Text),
    Notify("Please enter the rejection reason.", NotificationType.Error),
    Patch('TV Requests', varSelectedRequest, { 'Rejection / Cancellation Reason': txtReviewComment.Text });
    Set(varNewStatus, 'TVR Status'.Rejected);
    Set(varRemarks, "Rejected: " & txtReviewComment.Text);
    Select(btnDoStatusChange)
)
```

Requester can see "Required By" in red if late:

```
lblRequiredBy.Color = If(varSelectedRequest.'Required By Date' < Today() && varSelectedRequest.'Request
Status' <> 'TVR Status'.Closed, Color.Red, Color.Black)
```

9. Screen 4 – APPG Queue (scrAPPGQueue)

```
// galQueue.Items
SortByColumns(
    Filter('TV Requests',
        'Request Status' in ['TVR Status'.Submitted, 'TVR Status'.Under APPG Review', 'TVR
        Status'.Internal Alignment 1',
        'TVR Status'.Internal Alignment 2', 'TVR Status'.Funding Approved', 'TVR Status'.In
        Execution',
        'TVR Status'.Execution Completed'],
        chkMyItems.Value = false || 'APPG Owner'.Primary Email = User().Email,
        "tvr_statuschangedon", SortOrder.Ascending) // oldest waiting first
```

Show "Days waiting": `RoundDown(DateDiff(ThisItem.'Status Changed On', Now(), TimeUnit.Hours) / 24, 0)`.

Assign owner: combo box `cmbAPPGOwner` (Items = Users filtered by APPG team) → `Patch('TV Requests', varSelectedRequest, { 'APPG Owner': cmbAPPGOwner.Selected })`.

10. Screen 5 – Feasibility (scrFeasibility)

Gallery of options for the request + "Add Option" form.

```
// galOptions.Items
Filter('Feasibility Options', 'TV Request'.TV Request = varSelectedRequest.TV Request')

// "Select this option" button inside gallery
ForAll(Filter('Feasibility Options', 'TV Request'.TV Request = varSelectedRequest.TV Request) As o,
    Patch('Feasibility Options', o, { 'Is Selected': false }));
Patch('Feasibility Options', ThisItem, { 'Is Selected': true, Recommendation: 'Recommendation (Feasibility
Options)'.Recommended });
Set(varSelectedRequest, Patch('TV Requests', varSelectedRequest, { 'Selected Option': ThisItem, 'Estimated
Budget': ThisItem.'Quoted Cost' }));
```

(Here `'TV Request'` inside `'TV Request'.TV Request'` is the GUID primary key column of the table.)

Sample comparison shown to APPG:

Option	Institution	Capability	Cost	Weeks	Constraint	Recommendation
A	Stanford Nanofab Lab	5	39,500	10	Lab free from Nov	✓ Recommended
B	Fraunhofer IPMS	4	35,000	16	Shipping to Germany	Alternative
C	Internal Lab Fremont	3	22,000	20	Tool booked till Jan	Not recommended

A small column chart (Cost vs Weeks) helps decision. Quotes are uploaded with F09 into **03-Feasibility & Quotes**.

11. Screen 6 – Funding (**scrFunding**)

- Shows Cost Center, its Owner (from master), Selected option cost, approval history gallery.
- Send for Funding Approval** button: sets status Pending Funding Approval → flow **F04** creates the approval for the CCO (in Outlook + Teams + Approvals app).
- CCO can also open the request from the email link and see everything read-only.

```
// lblCCOwner.Text
varSelectedRequest.'Cost Center'.'Cost Center Owner'.'Full Name'
// galFundingHistory.Items
SortByColumns(Filter('Funding Approvals', 'TV Request'.'TV Request' = varSelectedRequest.'TV Request'),
    "createdon", SortOrder.Descending)
```

12. Screen 7 – Execution (**scrExecution**)

Three sections:

- Milestones** – editable gallery (Planned/Actual dates, % complete). Overall progress:

```
// when a milestone is saved:
Patch('TV Requests', varSelectedRequest,
    { 'Overall Progress %': Round(Average(Filter('Milestones', 'TV Request'.'TV Request' =
        varSelectedRequest.'TV Request'), '% Complete'), 0) })
```

- Risks / Issues / Exceptions** – add & update (severity High → flow F02 sends exception email).
- Costs** – add cost entries; show **Actual Total Cost** vs **Approved Amount** and utilization %:

```
lblUtil.Text = Text(varSelectedRequest.'Actual Total Cost' / varSelectedRequest.'Approved Amount', "0%")
```

Show it in red if > 100%. Plus "Expected Completion Date" date picker and "Add update / comment" box.

13. Screen 8 – Closure (**scrClosure**)

Fields: Final Outcome*, Lessons Learned*, Reuse Recommendation*, Final Report upload* (goes to **06-Final Report & Closure**), Final cost (read only, rollout).

```
// btnCloseRequest.OnSelect
If(
    IsBlank(txtOutcome.Text) || IsBlank(txtLessons.Text) || IsBlank(ddReuse.Selected),
        Notify("Outcome, lessons learned and reuse recommendation are mandatory.", NotificationType.Error),
        IsEmpty(Filter('Request Documents', 'TV Request'.'TV Request' = varSelectedRequest.'TV Request',
            'Document Type' = 'Document Type (Request Documents)'.Final Report)),
            Notify("Please upload the final report first.", NotificationType.Error),
            Patch('TV Requests', varSelectedRequest,
                { 'Final Outcome': txtOutcome.Text, 'Lessons Learned': txtLessons.Text,
                    'Reuse Recommendation': ddReuse.Selected.Value, 'Closed On': Now() });
                Set(varNewStatus, 'TVR Status'.Closed); Set(varRemarks, "Request closed");
                Select(btnDoStatusChange)
            )
```

14. Export to PDF

`btnExportPdf.OnSelect`:

```
Set(varPdf, 'F06-GeneratePDF'.Run(varSelectedRequest.TVID));  
Launch(varPdf.fileurl)
```

Flow F06 builds the PDF, saves it in SharePoint and returns the link.

15. Error handling rules (Validation & Error Handling)

- Every `Patch` wrapped with `IfError(..., Notify(FirstError.Message, NotificationType.Error))`.
- Disable buttons while saving: `btnSubmit.DisplayMode = If(varSaving, DisplayMode.Disabled, DisplayMode.Edit)`.
- Show a spinner when `varSaving = true`.
- Use `App.OnError` to log unexpected errors:

```
// App.OnError  
Notify("Something went wrong. Please try again or contact support.", NotificationType.Error)
```

16. Share the app

- Solution → app → **Share** with the Entra groups (`SG-TVRMS-Users`) – **not** individuals.
- Copy the app play URL into the environment variable `tvr_AppUrl` (used in email links).